

CSW + Cisco Secure Firewall Integration Guide

NetFlow (NSEL) visibility and FMC policy enforcement — customer reference

Cisco Secure Workload User Education
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Cisco Secure Workload + Cisco Secure Firewall

NetFlow (NSEL) ingestion and policy enforcement on the firewall

Disclaimer: Companion learning material — not official Cisco product documentation. Validate design, supported versions, and limits against your tenant, release notes, and:

- [FMC Integration Guide \(Cisco.com\)](#)
- [Secure Workload & Firewall deep dive \(secure.cisco.com\)](#)

Architecture diagrams: ARCHITECTURE.md only — do not duplicate diagrams in other docs.

Scope / query / ACP mapping: SCOPE-ACP-QUERY-MAPPING.md — scope queries, inventory filters, WorkloadObj__ dynamic objects, Jason Maynard + Jorge Quintero video paths.

Jorge Quintero — 3-part video series

Cisco Technical Marketing Engineer **Jorge Quintero** ([Cisco Secure Workload YouTube channel](#)) recorded the primary integration walkthrough. Watch **before** implementing.

Part	Title	Topics covered	Watch
1	Secure Workload & Firewall Integration	Introduction, design, high-level architecture, use cases	YouTube
2	Secure Workload & Firewall Integration	Deployment patterns, NSEL ingest, FMC connector setup, policy flow	YouTube

Part	Title	Topics covered	Watch
3	Secure Workload & Firewall Integration	Enforcement, telemetry validation, operations, troubleshooting	YouTube

Follow-up: [Secure Workload & Secure Firewall Integration Updates \(2025–2026\)](#) — latest behavior on the same channel.

Related Cisco Live session: Jorge Quintero — *Solving the Segmentation Puzzle with Secure Workload* ([BRKSEC-2161](#)).

Executive summary

In hybrid multi-cloud environments, policy controls often fragment across host firewalls, network firewalls, SDN, and cloud security groups — managed by different teams. **Cisco Secure Workload (CSW)** and **Cisco Secure Firewall** together provide a unified microsegmentation path for workloads where **host agents cannot be installed**, while still supporting mixed agent + agentless estates.

Three integrated value streams:

Use case	What it does	Primary path
Microsegmentation	Discover agentless flows (NSEL), model policy (ADM), enforce on FTD via FMC	Secure Firewall Connector + FMC Connector
Virtual Patch	Export workload CVEs to FMC; apply compensating IPS rules until patches ship	FMC Connector (Virtual Patch tab) + agents
Rapid Threat Containment	FMC detects anomaly → Remediation Module → CSW quarantine guardrails	FMC Remediation Module + CSW labels

Solution overview

CSW delivers three platform capabilities that intersect with Secure Firewall:

1. **Zero Trust Microsegmentation** — agent and agentless discovery, ADM-based policy, enforcement on host firewalls, DPUs, **network firewalls**, load balancers, and cloud controls.

2. **Vulnerability Detection and Protection** — runtime CVE visibility from agents; **virtual patch via Secure Firewall IPS**.
 3. **Behavioral Detection and Protection** — process/forensic analytics; **Rapid Threat Containment** with FMC correlation + CSW guardrails.
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Architecture

See [ARCHITECTURE.md](#) for all customer-facing diagrams (high-level integration, NSEL/FMC sequences, scope mapping, insertion options, virtual patch, rapid threat containment). This guide covers steps and design detail only — no duplicate diagrams here.

Two connectors — do not mix them

Integration	Connector	Purpose	Data direction
Flow visibility (NetFlow / NSEL)	Secure Firewall Connector on Ingest appliance	CSW sees flows from FTD/ASA without agents	Firewall → Ingest → CSW
Policy enforcement	FMC Connector (Cisco Secure Firewall in UI)	CSW pushes segmentation to FTD via FMC	CSW → FMC → FTD

Also available	Use when
NetFlow connector	Switches/routers export NetFlow v9/IPFIX — not FTD NSEL
Host agents	Process-level visibility + host enforcement
ACI connector	Fabric policy — separate from this FTD/FMC guide

Workload protection level (design first)

Before choosing firewall insertion mode, define the **security/trust boundary** your organization uses (subnet, zone, app tier, persona). The white paper recommends aligning network engineers and app owners on a measurable protection level — commonly **subnet-level** for agentless FTD insertion.

Persona	Typical boundary	Agent vs agentless
App / DevOps	App tier, namespace	Often agent-based
Network / Firewall	Subnet, zone, VRF	Often agentless (this guide)
Cloud	VPC/VNet, security group	Mixed

This drives whether you use **L2 transparent** (fine-grained), **L3 routed** (faster zone segmentation), or **cloud hub** patterns — see [Insertion options](#).

Prerequisites checklist

CSW + Ingest (visibility)

#	Requirement	Notes
1	CSW tenant (SaaS or on-prem) with admin access	Scopes and workspaces ready
2	Secure Workload Ingest virtual appliance	Active before enabling connector
3	Secure Connector (SaaS) or network path	Mandatory for SaaS
4	FTD/ASA can reach Ingest on UDP 4729	Copy IP:port from connector details
5	Capacity	45k fps per connector; 135k fps per Ingest appliance (on-prem)

FMC (enforcement)

#	Requirement	Notes
1	FMC or cdFMC with supported FTD devices	FTD managed by FMC only
2	FMC REST API user with administrative privileges	Required for connector
3	FTD associated with FMC; ACP assigned; deploy verified	Test FMC → FTD deploy before CSW
4	Secure Connector if SaaS CSW cannot reach FMC directly	HTTPS 443 to FMC
5	FMC HA: include standby FMC IP in connector	Auto-switch on failover
6	Supported versions	Segmentation: FMC 7.0.x+ (dynamic objects 7.0.1+); Virtual Patch: 7.2+

Virtual Patch (optional)

- CSW **agents** on target workloads
 - Threat intelligence **CVE data pack** on CSW cluster
 - Workloads use **FTD as default gateway** (traffic through IPS path)
 - FMC network discovery sees hosts under **Analysis** → **Network Map**
-

Part A — NetFlow / NSEL visibility (Secure Firewall Connector)

Goal: CSW receives bidirectional, stateful flow records from the firewall **without** host agents.

Step A1 — Plan ingest placement

1. **Manage** → **Workloads** → **Virtual Appliances** — confirm Ingest is **Active**.
2. Record **connector slot IP** and port (**4729** default).
3. Open firewall path: **UDP 4729** from FTD/ASA to Ingest.
4. **VRF: Manage** → **Agents** → **Configuration** → **Agent Remote VRF Configurations** — one connector per VRF.

Videos: [Connector Overview](#) · [Connector Deployment](#)

Step A2 — Enable Secure Firewall Connector in CSW

1. **Manage** → **Workloads** → **Connectors**.
2. Select **Cisco Secure Firewall Connector** (formerly ASA connector).
3. **Enable** on **Ingest appliance** (not Edge-only).
4. Record connector **ID** and **IP:port** from connector details page.

Limits: 1 Secure Firewall connector per Ingest appliance; 10 per tenant.

Step A3 — Enable NSEL on Secure Firewall

ASA example (replace <INGEST_CONNECTOR_IP>):

```
flow-export destination outside <INGEST_CONNECTOR_IP> 4729
flow-export template timeout-rate 1
!
policy-map flow_export_policy
  class class-default
    flow-export event-type flow-create destination <INGEST_CONNECTOR_IP>
    flow-export event-type flow-teardown destination
<INGEST_CONNECTOR_IP>
    flow-export event-type flow-denied destination <INGEST_CONNECTOR_IP>
    flow-export event-type flow-update destination <INGEST_CONNECTOR_IP>
    user-statistics accounting
service-policy flow_export_policy global
```

FTD: Configure NSEL export via FMC/device template — [ASA NetFlow Implementation Guide](#).

Videos (Jorge Quintero, 3-part series): [Part 1](#) · [Part 2](#) · [Part 3](#) · [2025–2026 updates](#)

Step A4 — Validate flow ingestion

1. Generate permitted and **denied** test traffic.
2. **Flow Analysis** — confirm firewall-sourced endpoints.
3. Connector heartbeat healthy under **Manage** → **Workloads** → **Connectors**.
4. Optional API filters: `user_src_NETFLOW_IDENTIFIED` / `user_dst_NETFLOW_IDENTIFIED`.

NSEL handling: flow-create/update → exported; flow-denied → **rejected** disposition; connector sends **forward** + **reverse** flows; **NAT flow-stitching** for end-to-end visibility.

Video: [Flow Analysis](#)

Step A5 — Label workloads and run discovery

1. Import **labels** from CMDB (ServiceNow), IPAM, or manual assignment.
2. Run **ADM** on scoped workspace.
3. Remain in **Monitor** mode until app owners validate dependencies.

Videos: [Basic Application Discovery](#) · [Enhancing Application Discovery](#) · [ADM & Policy Analysis](#)

Part B — Policy enforcement on Secure Firewall (FMC Connector)

Goal: CSW pushes **L3/L4 segmentation** to FTD via FMC using **Dynamic Objects** — inventory changes refresh membership without manual object edits.

CSW orchestrated rules are **L3/L4 east-west microsegmentation**. Layer-7 FMC features should not be added to CSW-controlled rules (not supported).

Step B1 — Onboard firewalls in FMC

1. FTD devices registered to **FMC** or **cdFMC**.
2. Each FTD assigned to an **Access Control Policy (ACP)**.
3. Verify FMC can **deploy** to FTD and traffic behaves as expected.
4. Document **one scope** → **one ACP** mapping plan (1:1 only).

Video: [FMC Integration \(Edge / Ingest / Appliance\)](#)

Step B2 — Create FMC Connector in CSW

1. **Manage** → **Workloads** → **Connectors** → **Firewall** → **Cisco Secure Firewall**.
2. Click **Configure your new connector here**.
3. Complete **New Connection** fields:

Field	Description
Connector Name	Unique name for this FMC connection

Field	Description
Username / Password	FMC REST API credentials (admin privileges)
CA Certificate	FMC CA cert for TLS validation, or Disable SSL on trusted networks
Server IP/FQDN and Port	FMC hostname or IP (443)
HTTP Proxy	If required: <proxy.host>:<proxy.port>
Secure Connector	Enable when SaaS CSW reaches FMC through Secure Connector

4. Click **Create**; verify **Event Log** tab for errors.

Official procedure: [FMC Integration Guide — Connection Settings](#)

Step B3 — Map scope to Access Control Policy

Prerequisite: Scope queries committed and inventory filters validated — [SCOPE-ACP-QUERY-MAPPING.md](#).

1. Open FMC connector → **Segmentation** tab → + **Add**.
2. **Add ACP Mapping:**
 - Select **Access Policy** (ACP) from FMC.
 - Map to exactly **one Scope**.
 - **Use Secure Workload Catch-All** — CSW catch-all in Default section, or use FMC default action.
 - **Enforcement Mode:**
 - **Merge** — CSW rules coexist with FMC rules; configure priority.
 - **Override** — CSW replaces user-created rules in mapped sections.
 - **Absolute / Default Policies priority** (Merge only):
 - Insert **above** or **below** existing Mandatory/Default rules per section.
3. Click **Submit**.

Scope mapping strategy

Model	When	Behavior
Child / leaf scope	Single application behind firewall	Push that app's policies + parent guardrails
Parent scope	Multiple applications	Push parent + eligible child policies (hierarchical)

Topology awareness

CSW pushes rules only to FTD devices on the **traffic path** for the mapped scope — avoids rule sprawl across unrelated firewalls.

Videos: [Policy Enforcement Overview](#) · [Where to Enforce](#) · [Policy Ordering](#)

Step B4 — What CSW creates in FMC

When enforcement is enabled on a workspace, CSW converts policies to FMC rules:

FMC artifact	Prefix	ACP section	Purpose
Golden rules	Workload_golden__	Mandatory	CSW ↔ agent communication behind FW
Segmentation rules	Workload__	Mandatory / Default	Enforced workspace policies
Catch-all rules	Workload_ca__	Default	If Use Secure Workload Catch-All enabled
Dynamic objects	Workload0bj__	Objects	Scope/inventory IP membership

CSW policy type	FMC placement
Absolute	Mandatory category
Default	Default category

Do not manually edit Workload_* rules in FMC — CSW restores them on next push. Independent FMC rules are safe in **Merge** mode if they avoid these prefixes.

Step B5 — Model policy, then enforce

1. Build or accept **ADM-recommended** policies in workspace tied to scope.
2. **Simulate / Monitor** — Policy Visual and flow analysis.
3. Change window: **enable enforcement** on workspace.
4. Confirm FMC deployment succeeds; verify Dynamic Objects update as inventory changes.

Videos: [Policy Lifecycle](#) · [Policy Validation and Analysis](#)

Step B6 — Validate enforcement

Test	Expected result
Allowed business flow	Passes; CSW/FMC show permit
Intentionally denied flow	Blocked at FTD; CSW rejected-flow evidence
New workload in scope	Workload0bj__ membership updates
FMC deployment	Automatic on CSW policy change
Compliance monitoring	Alerts on rejected flows / policy

Test	Expected result
	deviation
Check Event Log on FMC connector for push errors (Information / Warning / Error).	

Part C — Virtual Patch (optional)

Compensating **IPS** control for CVEs until permanent patches are deployed.

1. FMC connector → **Virtual Patch** tab → **Create a Virtual Patching Rule**.
2. Define **Host Query** (scope/filter) and **CVE Query** (CVSS, Cisco Risk Score, etc.).
3. CSW exports matching CVEs to FMC **Third-Party Vulnerabilities**.
4. FMC admin runs **Cisco Recommended Rules** to map CVEs → Snort signatures.
5. Apply IPS policy on ACP rule for affected traffic flows.

Prerequisites: Agents on workloads; CVE data pack; FTD as gateway; FMC 7.2+.

Part D — Rapid Threat Containment (optional)

Automated quarantine when FMC detects malicious behavior:

1. Install **FMC Remediation Module for Secure Workload** on FMC.
2. Configure CSW cluster IP, root scope, API key (**User Data Upload** permission).
3. Define **Correlation Rules** and **Correlation Policy** responses.
4. On trigger, Remediation Module sends affected IPs to CSW via API.
5. CSW **guardrail policies** (quarantine labels) enforce across agents and FTD.

Firewall insertion options

Mode	Segmentation	NSEL visibility	Best fit
L2 Transparent	Intra- + inter-subnet	Full	Legacy OS, fine-grained, localized
L3 Routed	Inter-subnet only	Inter-subnet only	Quick zone segment, dev/non-prod
ACI Service Graph PBR	Intra + inter EPG/ESG	Full (FW in path)	ACI + FTD multi-policy
AWS hub VPC	Inter-VPC	VPC logs + NSEL	Centralized cloud east-west
AWS distributed	Intra-VPC	VPC logs + NSEL	Per-VPC enforcement

Mode	Segmentation	NSEL visibility	Best fit
Azure hub VNet	Intra- + inter-VNet	NSG logs + NSEL	UDR steering
GCP hub VPC	Inter-VPC	VPC logs + NSEL	Centralized GCP

All patterns support **dual management**: CSW east-west + FMC north-south on the same ACP (Merge mode).

FAQs (from Cisco white paper)

Question	Answer
Mixed agent + agentless behind same mapped firewall?	Works. Agent-based rules may also appear in ACP (hierarchical policy). Primary use case remains agentless protection.
Map multiple scopes to one ACP?	No — strictly 1 scope : 1 ACP .
Add Layer-7 / FMC features to CSW rules?	Not supported — CSW rules are L3/L4 east-west only.
Dual-management (CSW + FMC rules)?	Merge mode + rule ordering; preserve order after go-live.
Modify a CSW-owned rule in FMC?	CSW overwrites on next push.
Supported FMC versions?	Dynamic objects: 7.0.1+ ; Virtual Patch: 7.2+ ; qualified: 7.0.1, 7.2.5 .

Recommended video learning path

Order	Video	Presenter / channel	Link
1	Secure Workload & Firewall Integration — Part 1	Jorge Quintero · @ciscosecureworkload	Watch
2	Part 2 — deployment & policy flow	Jorge Quintero	Watch
3	Part 3 — enforcement & operations	Jorge Quintero	Watch
4	Integration Updates (2025—	Cisco Secure Workload	Watch

Order	Video	Presenter / channel	Link
	2026)	channel	
5	Connector Overview	Cisco Secure Workload channel	Watch
6	Connector Deployment	Cisco Secure Workload channel	Watch
7	FMC + Edge / Ingest / Appliance	Cisco Secure Workload channel	Watch
8	Where to Enforce	Cisco Secure Workload channel	Watch
9	Policy Enforcement Overview	Cisco Secure Workload channel	Watch

NetFlow connector vs Secure Firewall connector vs API flowsearch

Method	Use when	Scale notes
Secure Firewall Connector (NSEL)	FTD/ASA agentless visibility	45k fps per connector; 135k fps per Ingest
NetFlow connector	Switches/routers (v9/IPFIX)	~15k fps per connector
Host agents	Process-level + host enforce	Best fidelity
OpenAPI flowsearch	Query ingested flows	Paginate; ≤24h; 1h chunks — not bulk ingest

POV evidence checklist

Evidence	Proves
Connector enabled + heartbeat healthy	Ingest path live
Sample flows in Flow Analysis (NSEL)	Visibility
ADM map for scoped app	Discovery value
FMC connector connected + Event Log clean	Enforcement path

Evidence	Proves
Scope ↔ ACP mapping screenshot	Topology binding
Workload__ / WorkloadObj__ in FMC	Policy push working
Monitor-mode policy stats	Safe pre-enforce review
Post-enforce deny test + FMC hit	Enforcement works
Allowed transaction test	No business break

Common pitfalls

Pitfall	Fix
Confusing NetFlow connector with Secure Firewall connector	NSEL from FTD/ASA → Secure Firewall Connector only
Expecting NSEL alone to enforce	Requires FMC connector + workspace enforce
Editing Workload_* rules in FMC	CSW will overwrite — change policy in CSW
Multiple scopes on one ACP	Redesign — 1:1 mapping only
L3 routed FW for intra-subnet segment	Use L2 transparent or host agents
Bulk flowsearch API timeouts	Paginate; chunk time windows
Skipping Secure Connector on SaaS	Required for Ingest and FMC reachability

Official Cisco documentation

Document	URL
FMC Integration Guide	cisco-secure-workload-and-fmc-integration-guide.html
Deep dive white paper	secure-workload-whitepaper-m-connectors.html
CSW 4.0 SaaS — Connectors	sec-workload-firewall-wp.html
Marketing white paper	ASA NetFlow Implementation Guide
ASA NSEL configuration	

Related repos

- [CSW-Secure-Firewall-Integration-Guide](#) — **this repo**
- [CSW-User-Education](#) — video library and onboarding
- [CSW-Policy-Lifecycle](#) — ADM → Monitor → Enforce
- [CSW-Agent-Installation-Guide](#) — host agents

- [CSW-Identity-Integration-Guide](#) — AD / Entra identity
- [CSW-ServiceNow-Connector-Guide](#) — CMDB labels
- [csw-splunk-integration](#) — SIEM
- [CSW-Compliance-Mapping](#) — compliance frameworks
- [csw-logs-check](#) — host enforce timing from agent logs

CSW + Secure Firewall — Architecture Reference

Single source of truth for all architecture diagrams in this repo. Other documents link here instead of repeating diagrams.

Video walkthrough: Cisco TME **Jorge Quintero** — 3-part series on [@ciscosecureworkload](#): [Part 1 — design](#) · [Part 2 — deployment](#) · [Part 3 — operations](#)

Derived from the [Cisco Secure Workload & Secure Firewall deep dive](#) and the [FMC Integration Guide](#).

Disclaimer: Companion material — validate against your CSW and FMC release notes before production.

1. High-level integration (microsegmentation)

This is the reference architecture for **agentless east-west segmentation** using NSEL visibility and FMC enforcement.

flowchart TB

```

subgraph Sources["Context & telemetry"]
    CMDB["CMDB / ServiceNow"]
    IPAM["IPAM / DNS"]
    Labels["Manual labels"]
end

subgraph Workloads["Application workloads"]
    Agent["Agent-based workloads<br/>host firewall enforce"]
    Agentless["Agentless workloads<br/>legacy / appliance / no agent"]
end

subgraph Visibility["Visibility path – NSEL"]
    FTD["Secure Firewall FTD / ASA<br/>Transparent or Routed mode"]
    NSEL["NSEL export<br/>NetFlow v9 · UDP 4729"]
    Ingest["Secure Workload Ingest Appliance"]
    SFC["Secure Firewall Connector<br/>Docker · up to 45k fps each"]
    FTD -->|"flow-create · teardown · deny · update"| NSEL
    NSEL --> SFC
    SFC --> Ingest
end

subgraph CSW["Cisco Secure Workload"]

```

```

FlowDB["Flow analysis & inventory"]
ADM["Application Dependency Mapping"]
Policy["Policy engine<br/>Monitor → Simulate → Enforce"]
Ingest --> FlowDB
CMDB --> FlowDB
IPAM --> FlowDB
Labels --> FlowDB
FlowDB --> ADM --> Policy
end

subgraph Enforcement["Enforcement path – FMC"]
  FMConn["FMC Connector<br/>HTTPS REST API :443"]
  FMC["Secure Firewall Management Center"]
  ACP["Access Control Policy<br/>Dynamic Objects"]
  Policy -->|"Scope ↔ ACP mapping<br/>Topology-aware rules"| FMConn
  FMConn --> FMC --> ACP
  ACP -->|"auto-deploy"| FTD
end

Agentless -->|"traffic through FW"| FTD
Agent -->|"optional: rules also pushed to ACP"| Policy

```

Key idea: Visibility and enforcement use **different connectors**. NSEL tells CSW what flows exist; the FMC connector pushes **L3/L4 segmentation policy** to FTD when you enforce a workspace.

2. Data flow — visibility (NSEL)

```

sequenceDiagram
    participant W as Workload traffic
    participant FW as Secure Firewall FTD/ASA
    participant IC as Secure Firewall Connector
    participant CSW as Secure Workload

    W->>FW: Application flow
    FW->>FW: Stateful inspection
    FW->>IC: NSEL record (create/update/deny/teardown)
    Note over IC: Decapsulate NetFlow v9<br/>Forward + reverse
    flows<br/>NAT flow-stitching
    IC->>CSW: Flow observations (agentless)
    CSW->>CSW: ADM · labels · policy modeling

```

NSEL capability	Customer value
Stateful events	See permits and denies — not just sampled NetFlow
Bidirectional flows	Connector exports forward + reverse observations

NSEL capability	Customer value
Flow stitching	End-to-end visibility when NAT is in path
Agentless discovery	No agent on legacy OS, appliances, or restricted hosts

Scale (on-prem Ingest, per Cisco white paper):

Metric	Limit
Per Secure Firewall connector	Up to 45,000 fps
Per Ingest appliance (total)	Up to 135,000 fps
Connectors per Ingest appliance	1 Secure Firewall connector
Connectors per tenant	10

3. Data flow — enforcement (FMC)

sequenceDiagram

```

participant Op as Operator
participant CSW as Secure Workload
participant FMC as FMC / cdFMC
participant FTD as FTD cluster

```

```
Op->>CSW: Enable enforcement on workspace
```

```
CSW->>CSW: Resolve scope inventory → IP sets
```

```
CSW->>FMC: Create/update Dynamic Objects (WorkloadObj_*)
```

```
CSW->>FMC: Push ACP rules (Workload_* · Workload_golden_* ·
```

```
Workload_ca_*)
```

```
FMC->>FTD: Deploy policy (no manual redeploy for CSW changes)
```

```
Note over FTD: Topology awareness – rules only on firewalls in traffic path
```

FMC rule prefixes (do not edit in FMC)

Prefix	Purpose	ACP section
Workload_golden__	Allow CSW ↔ agents behind the firewall	Mandatory
Workload__	Segmentation rules from enforced workspaces	Mandatory / Default
Workload_ca__	Catch-all rules (if Use Secure Workload Catch-All enabled)	Default
WorkloadObj__	Dynamic objects for scope/inventory membership	Objects

CSW **overwrites** changes to rules using these prefixes on the next push. Independent FMC rules are preserved in **Merge** mode if they do not use these prefixes.

4. Scope ↔ ACP mapping models

```
flowchart LR
    subgraph Single["Single application"]
        CS1["Child / leaf scope"]
        ACP1["One ACP"]
        CS1 -->|"1:1 mapping"| ACP1
    end

    subgraph Multi["Multiple applications"]
        Parent["Parent scope"]
        C1["Child scope A"]
        C2["Child scope B"]
        ACP2["One ACP"]
        Parent --> C1
        Parent --> C2
        Parent -->|"hierarchical policy"| ACP2
    end
```

Mapping	When to use	What CSW pushes
Child scope → ACP	One application protected by this firewall pair	Policies for that app + parent guardrails
Parent scope → ACP	Multiple apps behind same firewall estate	Parent + eligible child scope policies

Constraint: One ACP maps to **one** scope only (1:1). Plan ACP boundaries accordingly.

5. ACP mapping options

Setting	Merge mode	Override mode
Existing FMC rules	Kept; CSW rules inserted top or bottom	Replaced by CSW rules
Rule priority	Configurable: above/below Mandatory and Default sections	N/A
Catch-all	Optional: CSW catch-all or FMC default action	Same
Dual management	Yes — network team keeps north-south / baseline rules	CSW-only in mapped sections

Policy type placement:

CSW policy type	FMC ACP section
Absolute policies	Mandatory category
Default policies	Default category

6. Firewall insertion options (on-premises)

Choose insertion based on **protection level** (typically subnet / zone boundary) and segmentation granularity.

flowchart TB

```
subgraph L2["Layer 2 – Transparent / bump-in-wire"]
    L2W["Fine-grained · intra + inter-subnet<br/>Full NSEL visibility<br/>Legacy OS / localized workloads"]
end

subgraph L3["Layer 3 – Routed / gateway"]
    L3W["Faster time-to-segment · inter-subnet only<br/>Partial NSEL (no intra-subnet)<br/>Dev / non-prod zones"]
end

subgraph ACI["ACI + Service Graph"]
    ACIW["Intra + inter EPG/ESG<br/>PBR redirect or go-to/through<br/>CSW + FMC + ACI policies"]
end
```

Insertion	Segmentation	NSEL visibility	Best for
L2 Transparent	Intra- + inter-subnet	Full (intra + inter)	Fine-grained, legacy OS, localized apps
L3 Routed	Inter-subnet only	Inter-subnet only	Quick zone segmentation, distributed workloads
ACI Service Graph (PBR)	Intra + inter EPG/ESG	Full with FW in path	ACI fabric + FTD service graph
ACI Go-To / Go-Through	Inter or intra+inter	Varies by mode	North-south or strict in-path designs

7. Cloud insertion options (summary)

Platform	Pattern	East-west protection	Visibility
AWS	Centralized hub VPC FTD	Inter-VPC / inter-subnet	VPC flow logs + NSEL
AWS	Distributed per-VPC FTD	Intra-VPC inter-subnet	VPC flow logs + NSEL
Azure	Hub VNet FTD + UDR	Intra- + inter-VNet	NSG flow logs + NSEL
GCP	Hub VPC FTD	Inter-VPC	VPC flow logs + NSEL

All cloud patterns support **FMC dual management**: CSW owns east-west microsegmentation; FMC owns north-south ingress/egress.

8. Extended use cases

Virtual Patch (agent + FMC)

flowchart LR

```
Agent["CSW agent on workload"] -->|"CVE inventory"| CSW
CSW -->|"FMC Connector Virtual Patch tab"| FMC
FMC -->|"Third-Party Vulnerabilities + Snort mapping"| IPS["IPS
policy / Cisco Recommended Rules"]
IPS --> FTD
```

- Requires **agents** on workloads where virtual patch applies.
- Workloads must use an **FTD address as default gateway** for traffic to hit IPS.
- FMC 7.2+ for virtual patch; CSW publishes CVEs → FMC → fine-tuned IPS rules.

Rapid Threat Containment

flowchart LR

```
Anomaly["Malware / IPS / correlation event"] --> FMC
FMC -->|"Remediation Module API"| CSW
CSW -->|"Quarantine label / guardrail policy"| Enforce["Host agent +
FTD enforce"]
```

- FMC **Remediation Module** sends affected IPs to CSW.
 - CSW **guardrail policies** (label-based) quarantine workloads across enforcement points.
-

9. Connectivity requirements

Path	Protocol	Port	Notes
FTD/ASA →	UDP (NSEL)	4729 (default)	Firewall must

Path	Protocol	Port	Notes
Ingest			reach Ingest slot IP
CSW → FMC	HTTPS	443	REST API; admin credentials
SaaS CSW ↔ on-prem	Secure Connector	tunneled	Required for SaaS + on-prem FMC/Ingest
FMC HA	Include standby FMC IP	443	Auto-failover to new active FMC

10. Supported versions (reference)

Capability	CSW	FMC / FTD
Segmentation (ACP mapping, topology)	3.9.1.1+	7.0.x+ (dynamic objects: 7.0.1+)
Virtual Patch	3.8.1.1+	7.2+
Simplified segmentation workflow	3.9.1.1+	7.2 recommended
Engineering qualified (white paper)	—	7.0.1, 7.2.5

Platforms: FTD devices **managed by FMC only** (not standalone ASA for enforcement path). FTD **Transparent** and **Routed** modes supported.

Official references

Document	URL
Deep dive white paper	secure.cisco.com/secure-workload/docs/secure-workload-whitepaper
FMC integration guide	cisco-secure-workload-and-fmc-integration-guide.html
Marketing white paper	sec-workload-firewall-wp.html

Scope, Query, Inventory Filters & ACP Mapping

How **CSW scope queries**, **inventory filters**, and **clusters** become **FMC Dynamic Objects** (WorkloadObj__*) and **ACP rules** — with video and documentation references.

Disclaimer: Companion material. Authoritative sources:

- [FMC Integration Guide](#)

- [Secure Workload & Firewall deep dive](#)
- CSW User Guide — *Scopes, Inventory Filters, Manage Policy Lifecycle*

Architecture diagrams: [ARCHITECTURE.md](#) · Firewall integration steps: [INTEGRATION-GUIDE.md](#)

Terminology — do not confuse these

Concept	What it is	Role in firewall enforcement
Scope	Hierarchical container for policy (parent/child tree)	One scope maps to one FMC ACP (1:1)
Scope query	Boolean query defining which workloads belong in a scope	Drives scope membership; changes need commit + impact review
Label	Key/value metadata on workloads (CMDB, cloud tag, manual)	Used inside scope queries and policy consumers/providers
Inventory filter	Named, reusable query (scope-bound) for grouping workloads	Becomes IP sets → WorkloadObj__* dynamic objects in FMC
Cluster	ADM-derived workload group (often ephemeral)	Can be converted to inventory filter for stable enforcement
WorkloadObj__*	FMC Dynamic Object created by CSW FMC connector	Membership updates automatically when inventory changes
Workload__* / Workload_ca__*	FMC ACP rules pushed from enforced CSW workspaces	L3/L4 segmentation rules referencing WorkloadObj__ objects
CSDAC (FMC Dynamic Attribute Connector)	Separate FMC feature for cloud tag → FMC objects	AWS/Azure/GCP tags — not the CSW WorkloadObj__ path

Key FMC integration quote (Cisco docs):

Secure Workload enforced segmentation policies are converted into Access Control Policies, utilizing **IP address sets derived from scopes, inventory filters, and clusters**. These sets are transformed into **dynamic objects** within FMC.

Network inventory is **dynamically updated** from inventory filters; when workloads are added, changed, or removed, CSW updates FMC Dynamic Objects and **auto-deploys** to FTD — no manual FMC redeploy for object membership changes.

End-to-end mapping chain

flowchart LR

```
subgraph CSW["Cisco Secure Workload"]
  L["Labels<br/>CMDB · cloud · manual"]
  SQ["Scope query"]
  IF["Inventory filters<br/>+ clusters"]
  POL["Policies<br/>consumer / provider"]
  WS["Workspace<br/>Monitor → Enforce"]
  L --> SQ
  SQ --> IF
  IF --> POL
  POL --> WS
end

subgraph FMC["FMC / FTD"]
  MAP["ACP ↔ Scope mapping<br/>1:1 · topology-aware"]
  DO["Dynamic Objects<br/>WorkloadObj__*"]
  RULES["ACP rules<br/>Workload__* · Workload_golden__*"]
  WS -->|"FMC Connector"| MAP
  IF --> DO
  POL --> RULES
  DO --> RULES
  RULES --> FTD["FTD deploy"]
end
```

Order of operations: Labels → Scope tree + queries → Inventory filters → ADM/policy → **then** FMC connector ACP mapping → enforce workspace.

Phase 0 — Foundations (Jason Maynard)

Jason Maynard (*How Hard Can It Be?* series) covers CSW building blocks on his [YouTube channel](#). Complete **before** ACP mapping — Jorge Quintero's firewall series assumes you understand scopes, labels, and filters.

Order	Video	Steps / topics (from catalog)	Watch
1	Cisco Secure Workload: Scopes	Create scope tree; group workloads for policy; parent/child hierarchy	YouTube

Order	Video	Steps / topics (from catalog)	Watch
2	Cisco Secure Workload: Labels	Tag workloads; use labels in queries and policy	YouTube
3	Cisco Secure Workload: Inventory Filters	Build filters to focus inventory; reusable query groups	YouTube
4	Application Dependency Mapping & Policy Analysis	Map dependencies; derive segmentation policy from flows	YouTube
5	Dynamic Workloads & Policy	Policy adapts as workloads move or scale	YouTube
6	Policy Visual and Quick Analysis	Visualize policy impact before enforce	YouTube
7	Production and Test Risk Reduction	Macro-segment prod vs non-prod (common first firewall scope)	YouTube
8	Flow Analysis	Validate NSEL/agent flows before policy (pairs with Part A of firewall guide)	YouTube

Official channel refresh: [Inventory Filters \(2025–2026\)](#) — [@ciscosecureworkload](#)

Phase 1 — Scope query (CSW UI / API)

Define **which workloads** belong in each scope before FMC mapping.

Steps (CSW User Guide + OpenAPI)

Step	Action	Notes
1	Manage → Scopes (or tenant scope tree)	Plan parent (multi-app) vs child/leaf (single app) before FMC
2	Create a New Scope	Name aligned to app or

Step	Action	Notes
3	Edit scope query	zone boundary (e.g. Prod-Payments-App) Use labels, IP, hostname, agent attributes, connector metadata
4	Review Scope Query Change Impact	UI modal shows affected workloads and policies
5	Commit scope query changes	API: POST /scopes/{id}/commit_query_changes — until committed, query is draft
6	View workloads in scope	Confirm count matches expectation before enforce
7	Policy inheritance	Child scopes inherit parent guardrail policies (relevant for leaf → ACP mapping)

Scope mapping strategy for FMC

FMC mapping	Map this scope type	CSW pushes
Single application	Child / leaf scope	That app's policies + parent guardrails
Multiple applications	Parent scope	Parent + eligible child scope policies (hierarchical)

Constraint: One ACP ↔ one scope only. Plan one FMC Access Control Policy per segmentation boundary.

Phase 2 — Inventory filters & clusters

Inventory filters are the **primary query objects** that become **WorkloadObj__*** IP sets in FMC.

Steps (CSW User Guide)

Step	Action	Notes
1	Create an Inventory Filter	Scope-bound; name reflects consumer/provider role

Step	Action	Notes
		(e.g. web-tier, db-tier)
2	Validate an inventory filter query	API: validate before save — check match count
3	Review Filter Change Impact	Shows policy/workspace impact before commit
4	Run ADM / accept clusters	ADM may propose clusters as workload groups
5	Convert a Cluster to an Inventory Filter	Stabilize ADM output for enforcement
6	Use filters as policy consumer/provider	Policies reference filters, not raw IP lists
7	Dynamic Workloads & Policy	When labels/IPs change, filter membership refreshes → FMC objects update

Query building tips

- Combine **labels** (from ServiceNow, cloud connectors, manual) with **IP/subnet** for agentless NSEL-visible hosts.
- Keep filter names stable — FMC `WorkloadObj__` names derive from CSW objects.
- Prefer **inventory filters** over ad-hoc IP lists in policies for FMC dynamic object hygiene.

Phase 3 — Policy workspace (before FMC push)

Step	Action	Notes
1	Create workspace tied to target scope	Primary workspace for the app
2	ADM or manual policies — consumer/provider use inventory filters	See Jason Maynard ADM + Policy Visual videos
3	Monitor / Simulate	No FTD push yet
4	Policy Visual — allowed/denied impact	Validate with app owner
5	Document Absolute vs Default vs Catch-All rank	Maps to FMC Mandatory vs Default sections

Phase 4 — ACP ↔ Scope mapping (FMC Connector)

After scope queries, filters, and monitor-mode policy are validated — configure FMC connector **Segmentation** tab.

Prerequisites

- FMC connector **connected** (Event Log clean)
- FTD devices assigned to target **ACP**; FMC deploy verified
- **One scope** selected per ACP

Steps (FMC Integration Guide)

Step	UI action	Detail
1	Manage → Workloads → Connectors → Cisco Secure Firewall	Open existing FMC connection
2	Segmentation tab → + Add	Add ACP Mapping
3	Choose Access Policy (ACP) from FMC dropdown	Must match FTD assignment in FMC
4	Map to one Scope	1:1 — cannot map multiple scopes to one ACP
5	Use Secure Workload Catch-All	CSW catch-all in Default section, or FMC default action
6	Enforcement Mode	Merge (dual-management) or Override (CSW-only in section)
7	Absolute / Default Policies priority	Merge only: insert above/below Mandatory/Default rules
8	Submit	Topology awareness pushes rules only to FTDs on traffic path
9	Enable enforcement on CSW workspace	Triggers push of objects + rules
10	Verify in FMC	Objects: WorkloadObj__* . Rules: Workload__*, Workload_golden__*, Workload_ca__*

Step	UI action	Detail
11	Event Log tab	Filter errors (red); download JSON/CSV if needed

What CSW creates in FMC

FMC artifact	Prefix	Section
Golden rules (CSW ↔ agents behind FW)	Workload_golden__	Mandatory
Segmentation rules from enforced workspace	Workload__	Mandatory / Default
Catch-all (if enabled)	Workload_ca__	Default
Dynamic objects from scopes/filters/clusters	WorkloadObj__	Object Management

CSW policy rank	FMC ACP category
Absolute	Mandatory
Default	Default

Do not edit Workload_* rules or WorkloadObj__ objects manually in FMC — CSW overwrites on next push.

Edit / view mapping later

1. **Defend** → **Segmentation** → select policy → pencil icon → **Submit** → **Save**
2. **Domains** tab (FMC connector config) — view which FMC domains are enforced (3.9.1.1+)

Phase 5 — Firewall integration videos (Jorge Quintero)

After Phase 0–4 concepts, watch Jorge's series for **NSEL + FMC connector deployment**:

Part	Focus	Watch
1	Design & architecture	YouTube
2	Deployment & policy flow	YouTube
3	Enforcement & operations	YouTube

CSW WorkloadObj__ vs FMC CSDAC (Dynamic Attributes)

	CSW → FMC (WorkloadObj__)	FMC CSDAC
Source	CSW scopes, inventory filters, clusters	AWS/Azure/GCP/ VMware tags via FMC connector
Configured in	CSW UI + FMC connector Segmentation	FMC → Dynamic Attributes Connector
Use with CSW firewall integration	Yes — required for CSW-driven segmentation	Optional — parallel cloud-native object path
Doc	FMC Integration Guide	CSDAC overview

For a **CSW POV with FMC enforcement**, you need the **CSW FMC connector** path. CSDAC is complementary when FMC also needs cloud tag-based objects outside CSW policy.

Validation checklist

#	Check	Pass criteria
1	Scope query committed	Workload count stable; impact review done
2	Inventory filters validated	Match expected tiers; names stable
3	Policies use filters (not static IPs)	Consumer/provider reference filters/clusters
4	Workspace in Monitor	ADM/policy validated with app owner
5	ACP mapping 1:1	One scope per ACP; topology correct
6	FMC WorkloadObj__* populated	Objects show current IPs
7	Post-enforce deny test	Blocked at FTD; CSW shows rejected flow
8	Add/remove test workload	Filter membership + FMC object update without manual FMC edit

FAQs (scope / query / ACP)

Question	Answer
Multiple scopes on one ACP?	No — strictly 1:1
Mixed agent + agentless behind same FW?	Works; agent host rules may also appear in ACP (hierarchical policy)
Edit Workload__ rule in FMC?	CSW restores on next push
Layer 7 on CSW rules?	Not supported — L3/L4 east-west only
Scope query change after enforce?	Review impact → commit → FMC objects refresh on push
Cluster vs inventory filter?	Convert clusters to filters for stable FMC object mapping

Recommended learning order

Jason Maynard: Scopes → Labels → Inventory Filters → ADM → Policy Visual

- Jorge Quintero: Firewall Integration Parts 1–3
- This doc: Scope query → Filters → ACP mapping → Enforce
- INTEGRATION-GUIDE.md: NSEL ingest + FMC connector setup detail